Sean Luka Girgis

PySpark / SQL Data Engineer | ETL, Hadoop/Spark, Data Validation & Production Support

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[LinkedIn](#) | [GitHub](#) | [Portfolio](#)

Professional Summary

Senior Data Engineer with 20+ years of enterprise technology experience across Python, SQL, PySpark/Spark, ETL, high-volume data processing, relational databases, Hadoop/Hive concepts, data validation, reporting, unit/integration testing, and production issue resolution. Strong background designing maintainable data modules, optimizing SQL and data pipelines, validating large datasets, troubleshooting production issues, and supporting stakeholder-facing analytics. Best aligned to PySpark/SQL data engineering roles requiring ETL scripts, data extraction, transformation, validation, statistical reporting, and high-volume pipeline support; healthcare claims, Microsoft SQL Server, Impala, Hue, Oozie, Yarn, Sqoop, and Kafka are positioned as adjacent or learning areas rather than overstated production ownership.

Professional Experience

Senior Capacity & Data Engineer at CITI (2017-11 – 2025-12)

- Built Python/Pandas/PySpark ETL pipelines ingesting telemetry from 6,000+ infrastructure endpoints and transforming raw operational data into curated datasets for reporting, analytics, forecasting, and validation workflows.
- Designed AWS and relational data platform components using S3, AWS Glue, Athena, Redshift, Oracle, SQL, and PySpark to support scalable transformation, querying, and stakeholder-facing data products.
- Created and maintained ETL scripts, SQL queries, reporting datasets, validation checks, and analytics outputs used by technical and business teams for operational decision-making.
- Optimized SQL queries, Oracle schemas, Redshift-backed workloads, indexes, views, partitioning patterns, and historical retention models for performance and maintainability.
- Built data validation, reconciliation, and quality review practices across BMC TrueSight, CA Wily, AppDynamics, Oracle, and cloud-backed reporting layers to improve trust in pipeline outputs.
- Analyzed and resolved production data issues from internal customers by tracing source feeds, transformation logic, SQL behavior, and reporting-layer discrepancies.
- Prepared model-ready and statistical reporting datasets for Prophet and scikit-learn forecasting workflows, supporting capacity planning and bottleneck prediction.
- Documented requirements, detailed designs, transformation logic, operational runbooks, and support steps to improve maintainability and production handoff.

Performance Engineer at G6 Hospitality LLC (2017-03 – 2017-11)

- Developed analytics and reporting outputs from Dynatrace AppMon and Gomez Synthetic Monitoring data for Brand.com and critical hospitality systems.
- Led the FAST project, mining real-user and synthetic performance metrics to identify optimization opportunities and communicate findings to engineering teams.
- Built dashboards and operational reporting views that converted raw monitoring data into actionable performance and reliability insights.

- Supported production issue analysis by correlating monitoring signals, transaction behavior, and system performance data.

SME for CA APM (Senior Consultant) at CA Technologies / TIAA-CREF (2011 – 2016)

- Managed enterprise telemetry environments with 50+ Enterprise Managers and 4,000–6,000 instrumented agents, supporting monitoring, operational analytics, dashboards, and production reliability.
- Built Perl and Korn shell automation for extracting, transforming, and distributing performance data, replacing manual exports with repeatable reporting workflows.
- Designed dashboards, alert logic, SLA thresholds, technical documentation, and reusable reporting standards adopted across client environments.
- Partnered with J2EE, WebLogic, infrastructure, database, and operations teams to analyze data, troubleshoot bottlenecks, and resolve production issues.

Performance Test Engineer at AT&T (2010-08 – 2011-07)

- Analyzed enterprise telecom applications under load to identify throughput limits, SQL/JDBC bottlenecks, CPU, memory, thread, and garbage-collection constraints.
- Configured JMX Monitoring, Wily Introscope, and thread-dump automation to improve runtime visibility during performance and regression testing.
- Documented baseline metrics, test findings, regression thresholds, and release-readiness notes used by engineering teams for production decisions.

Senior Systems & Data Migration Engineer at Sabre (2008-05 – 2010)

- Led migration of a high-throughput shopping engine from 200+ MySQL nodes to a 6-node Oracle RAC cluster, preserving transaction performance and data integrity.
- Designed and optimized SQL and Oracle access patterns for high-volume workloads while reducing physical hardware footprint by 95%.
- Built C++/OCCI/OCI validation and regression testing utilities to verify correctness across millions of migrated transaction records.
- Automated schema conversion, data validation, and phased cutover support for a mission-critical enterprise migration.

Education

Post-Graduate Diploma in Computer Engineering Technology / Computer Science
Humber College, Toronto, Canada

Bachelor of Science in Civil Engineering
Zagazig University, Egypt

Skills

Languages: Python, SQL, PySpark, PL/SQL, XML, JSON, Perl, Korn Shell, C++, Java

Flagship Projects

HorizonScale — AI Capacity Forecasting Engine
Built a forecasting-oriented analytics workflow using Python, PySpark, Prophet, scikit-learn, and Streamlit to

process infrastructure telemetry, prepare model-ready datasets, validate pipeline outputs, and deliver capacity insights.
Technologies: Python, PySpark, SQL, Prophet, scikit-learn, Streamlit, Data Validation

Serverless Lakehouse Platform (AWS)
Designed a scalable AWS data platform pattern using S3, Glue, Athena, Redshift, PySpark, and Parquet for raw ingestion, transformation, curated analytics datasets, query optimization, and reporting access.
Technologies: AWS S3, AWS Glue, Athena, Redshift, PySpark, Parquet, SQL

AI-Powered Job Search Pipeline
Built a Python automation pipeline with semantic duplicate detection, LLM-based job scoring, JSON artifact generation, quality gates, document rendering, and application tracking using FAISS, sentence-transformers, and LLM APIs.
Technologies: Python, FAISS, Sentence Transformers, LLM APIs, JSON, Automation, Quality Gates